

Tokenizing Real Estate with Trust: A Dual-Approval and Compliance Framework on ERC1155C

1st Surjo Santosa Sastroharjono
Information System Management
Department
Bina Nusantara University
Jakarta, Indonesia 11480
surjo.sastroharjono@binus.ac.id

2nd Gunawan Wang
Information System Management
Department
Bina Nusantara University
Jakarta, Indonesia 11480
gwang@binus.edu

Abstract—Real estate tokenization has attracted global attention for its potential to increase liquidity, transparency, and retail participation. In Indonesia, however, the equivalent of REITs—*Dana Investasi Real Estat* (DIRE)—has struggled due to limited investor participation, high transaction costs, and illiquidity. Legal restrictions on land title transfer, which must be executed before a notary and registered at the National Land Agency (BPN), make direct asset-based tokenization infeasible. A more viable alternative is debt- or rights-based tokenization, where economic rights such as rental income or fixed returns are issued as digital tokens. This paper proposes an on-chain issuance framework that embeds compliance directly into smart contract design. The framework extends ERC1155 into ERC1155C, introducing dual approval by a notary and fund manager, as well as whitelist restrictions to enforce KYC/AML requirements. All issuance events are immutably logged, creating transparency and regulatory auditability. Supporting information is stored on IPFS to accompany the offering stage, reducing administrative costs and making debt-based tokenization competitive with conventional bank financing. By focusing on the issuance stage, this study highlights how compliance-by-design can strengthen trust and adoption of tokenized real estate in regulated markets such as Indonesia.

Keywords—Real estate tokenization, Blockchain compliance, Smart contracts, On-chain issuance; Indonesia

I. INTRODUCTION

Real estate is one of the largest asset classes in Indonesia, yet its potential for capital formation remains underdeveloped. The Indonesian REIT-equivalent, *Dana Investasi Real Estat* (DIRE), has shown limited growth due to low investor participation, high transaction costs, and illiquidity [1], [2]. A key barrier lies in the legal structure of property ownership. Under Indonesian law, transfer of land and building titles must be executed before a notary/land deed official (PPAT) and registered with the National Land Agency (BPN) [3], [4]. This requirement makes direct asset-based tokenization infeasible. A more realistic alternative is *rights/debt-based tokenization*, where economic rights such as rental income, revenue sharing, or fixed returns are represented by tokens rather than direct title transfer [5].

The financial context reinforces this approach. Interest rates on bank loans in Indonesia are typically 8–12% [6], while deposit rates remain only 4–6% [7], creating strong demand for debt-based instruments. However, traditional bond issuance involves high upfront costs and complex

procedures, making it unsuitable for small and medium enterprises (SMEs) [8].

While certain aspects of issuance and investment can technically be digitized through online bond platforms or securities crowdfunding portals, these remain centralized and constrained by regulatory bottlenecks [9]. Manual verification, centralized custody, and sequential filings sustain high costs and limit scalability [10]. Blockchain differs by embedding compliance directly into smart contracts, reducing reliance on intermediaries, enabling real-time transparency, and creating immutable audit trails accessible to all stakeholders [11]. Compared to conventional digital platforms, the blockchain-based issuance framework offers three advantages: (i) lower costs through reduced administrative overhead, (ii) enhanced trust via verifiable dual approval and whitelist restrictions, and (iii) broader inclusivity through fractional participation [12].

Within the broader landscape of digital property trading, tokenization represents not a replacement but an evolution of digital platforms. By providing a flexible, rights-based, and compliance-embedded issuance mechanism, tokenization bridges conventional regulation with blockchain-enabled efficiency [13]. To the best of our knowledge, this is the first study to propose an issuance framework that encodes Indonesian compliance requirements directly into smart contract design.

II. THEORETICAL FRAMEWORK

A. Tokenization Process

Real estate tokenization in Indonesia cannot be implemented through direct asset transfer due to strict land ownership regulations requiring PPAT and BPN registration [3], [4]. Instead, tokenization focuses on economic rights such as rental income, revenue sharing, or debt returns [5], [8].

Figure 1 illustrates the off-chain verification workflow where a property owner provides legal documents to a Special Purpose Vehicle (SPV). These documents are validated by a notary to ensure legal enforceability and by a licensed fund manager to check financial feasibility and investor protection. Based on this review, the SPV drafts a debt agreement, with underwriting responsibilities if the offering is undersubscribed [14].



Fig. 1. Off-Chain Pre-Approval Process

Once approved, the process shifts on-chain in a simplified workflow: the SPV submits a request, followed by approval from both the notary and the fund manager. Only after dual approval are the tokens minted. The subsequent flow mirrors conventional debt instruments—initial offering, escrow placement, fund disbursement to the owner, periodic return distribution, and contract settlement—but enhanced with blockchain-based transparency and auditability [13], [14].

B. Smart Contract Mechanism

The on-chain stage is implemented via EVM-compatible smart contracts using token standards such as ERC20, ERC721, and ERC1155. While ERC20 focuses on fungible tokens and ERC721 on unique non-fungible tokens, ERC1155 offers flexibility by combining multiple asset classes in a single contract. For real estate, this allows different rights (e.g., fixed income vs. revenue sharing) to be represented together [15], [16].

To adapt to Indonesia's regulatory context, we extend ERC1155 into ERC1155C (Compliance). This mechanism introduces (i) Dual approval functions: issuance requires validation by both a notary and a fund manager; (ii) Whitelist restrictions: ensuring only KYC/AML-verified investors can participate and (iii) Immutable event logs: each step—request, approval, and execution—is permanently recorded for regulatory audit [14].

This compliance-by-design approach integrates Indonesia's legal requirements directly into blockchain logic, ensuring that token issuance is not only technically secure but also legally enforceable [13].

III. RESEARCH METHOD

This study applies a blockchain architecture approach, focusing exclusively on the smart contract design as the core of the issuance framework. The objective is not to model the entire investment lifecycle, but to demonstrate how compliance requirements can be directly encoded into the tokenization process [16].

Unlike traditional real estate investment models that rely on centralized intermediaries for validation and enforcement, smart contracts allow programmable compliance. In this design, the issuance of real estate tokens cannot proceed without dual approval from a licensed notary and a fund manager. Moreover, a whitelist mechanism enforces KYC/AML restrictions to ensure only verified investors can participate [14].

By limiting the scope of analysis to the issuance smart contract, the paper highlights the critical foundation of a compliant tokenization system in Indonesia. This focus underscores how blockchain can embed transparency, security, and regulatory alignment at the technical core of the process [13].

A. Roles in the Issuance Process

As illustrated in Figure 2, the issuance process involves four interrelated actors: (i) Special Purpose Vehicle (SPV) – initiates on-chain requests to represent fractional economic rights; (ii) Notary – validates authenticity and enforceability of the documents; (iii) Fund Manager – reviews financial compliance, investor protection, and OJK alignment and (iv) Token Recipient (Property Owner) – receives financing through token issuance and commits to periodic payments (interest or revenue sharing) as stated in the agreement [5], [8].



Fig. 2. Roles in the Issuance Process

Together, these roles replicate conventional Indonesian real estate practices but are encoded into blockchain logic, embedding compliance directly into the issuance process [14].

B. Smart Contract Logic

The smart contract logic is structured around four main functions that define the issuance workflow:

- requestMint() – SPV submits issuance requests with property metadata (See Figure 3).
- approveByNotary() – records legal validation.
- approveByManager() – confirms financial compliance.
- executeMint() – finalizes issuance only after both approvals.

```
function requestMint(
  address to,
  uint256 id,
  uint256 amount,
  FeeInfo calldata fees,
  string calldata setUriIfEmpty,
  bytes32 docHash,
  string calldata docCid
) external {
  require(kyc[msg.sender] && kyc[to], "KYC required");
  require(to != address(0), "bad to");
  require(amount > 0, "amount=0");

  MintRequest storage r = mintRequests[mintRequestId];
  r.to = to;
  r.id = id;
  r.amount = amount;
  r.fees = fees;
  r.uri = setUriIfEmpty;
  r.docHash = docHash;
  r.docCid = docCid;

  emit MintRequested(mintRequestId, to, id, amount);
  mintRequestId++;
}
```

Fig. 3. Subset of requestMint() script

Supporting functions strengthen compliance and investor protection, such as:

- setWhitelist(address) – restricts token holding to verified participants.
- freezeAccount(address) – allows accounts to be blocked in case of suspicious or non-compliant activity.

All functions generate immutable on-chain events (MintRequested, ApprovedByNotary, ApprovedByManager, MintExecuted), which provide transparency and verifiability for regulators and investors [14], [15].

C. On-Chain Issuance Workflow

The on-chain issuance workflow is depicted in Figure 4. After the SPV submits a minting request (requestMint), the notary validates legality (approveByNotary). The fund manager then confirms financial oversight (approveByManager). Only when both approvals are securely recorded can the SPV execute executeMint, minting the tokens and logging the MintExecuted event. Each step emits immutable logs — MintRequested, ApprovedByNotary, ApprovedByManager, and MintExecuted—that together create a verifiable audit trail for regulators and investors.

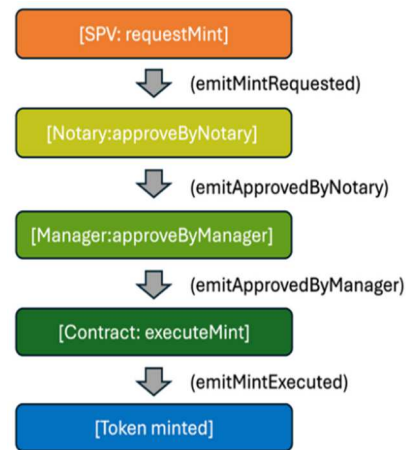


Fig. 4. On-chain issuance workflow for real estate tokenization

D. Validation

The methodology is validated through simulation of stakeholder interactions (SPV, Notary, Fund Manager, Investors) in a controlled blockchain environment. Evaluation criteria include: (i) Correctness – unauthorized or non-whitelisted accounts cannot mint or hold tokens; (ii) Compliance embedding – issuance cannot bypass dual approval or AML/KYC restrictions and (iii) Transparency – all issuance steps remain permanently auditable on-chain [1], [14], [15].

IV. RESULTS

The proposed ERC1155C smart contract was successfully implemented in Solidity (v0.8.19) and deployed on the Polygon Amoy testnet (address: `0x7c80E676758cc6f1748ddF0c02dB0abE8Ec42631`). Validation confirmed that the contract correctly enforced dual approval and whitelist restrictions, preventing unauthorized issuance and ensuring compliance alignment.

A. Test Results

Automated testing produced the following outcomes as shown in Table 1.

TABLE I. TEST CASE RESULTS

Test Case	Expected Result	Outcome
Mint without approval	Transaction reverted	Pass
Mint with only notary approval	Transaction reverted	Pass
Mint with only manager approval	Transaction reverted	Pass
Mint with dual approval	Token minted, event logged	Pass
Transfer to non-KYC address	Transaction reverted	Pass

These outcomes confirm that the contract consistently rejected unauthorized transactions, enforced dual approval before issuance, and maintained immutable on-chain records. The results demonstrate that the proposed framework is

technically feasible and operates as intended in a live blockchain environment [14], [15].

B. Deployment Outcomes

The ERC1155C smart contract was deployed on the Polygon Amoy testnet. The on-chain transaction history confirms that issuance events follow the intended compliance workflow.

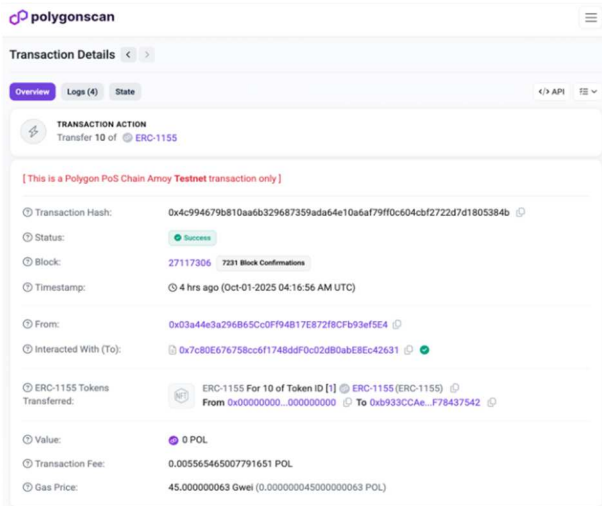


Fig. 5. Example On-Chain Result (Polygon Amoy transaction log)

The example in figure 6 shows sequential events:

- Tokens could only be minted after approvals from both the notary and the fund manager.
- Unauthorized or non-whitelisted addresses were rejected.
- Event logs (MintRequested, ApprovedByNotary, ApprovedByManager, MintExecuted) were consistently recorded on-chain, ensuring full auditability.

The full implementation and test scripts are publicly available for reproducibility via GitHub Repository (<https://github.com/ljcom/erc1155c>) and PolygonScan (<https://amoy.polygonscan.com/tx/0x4c994679b810aa6b329687359ada64e10a6af79ff0c604cbf2722d7d1805384b>).

V. DISCUSSION

A. Main Contribution

The primary contribution of this study is the development of a compliance-embedded smart-contract architecture that covers the complete lifecycle of tokenized economic rights—starting from issuance and extending into controlled trading and periodic revenue distribution. Unlike conventional models where compliance is executed manually or externally, this framework demonstrates that legal, financial, and KYC/AML requirements can be enforced as programmable logic within a unified contract system.

1) Issuance as Programmable Compliance

The ERC1155C framework strengthens the issuance process through three mechanisms: (i) Dual Approval Enforcement – Token minting is strictly conditioned on

sequential approvals from the notary and fund manager, representing Indonesia's legal and financial safeguards; (ii) Document Binding via IPFS – Token metadata incorporates hashed legal and financial documents, anchoring off-chain evidence to on-chain issuance events; and (iii) Immutable Audit Trail – All actions (MintRequested, ApprovedByNotary, ApprovedByManager, MintExecuted) become permanent regulatory records.

Collectively, the issuance phase restructures compliance from a post-hoc administrative activity into a deterministic protocol requirement, improving transparency and reducing operational cost.

2) Regulated Trading Through Restricted Marketplace Logic

A key extension beyond this design is the inclusion of a restricted, role-controlled marketplace, which allows compliant secondary trading without introducing a public or permissionless market. The design includes: (i) KYC-gated listing and purchasing through createListing and buyListing; (ii) Freeze controls (setFreezeAccount, setFreezeId) to support AML response, dispute handling, or regulatory intervention; and (iii) MockUSDT/ERC-20 payment layer, enabling a fiat-mirrored trading experience aligned with OJK and BI rules on IDR-denominated settlement [17].

This marketplace design demonstrates how token circulation can remain compliant even after issuance, enabling liquidity in a controlled environment while preventing unverified participants from entering the system.

3) Automated and Pro-Rata Interest / Revenue Distribution

The third dimension of contribution is the inclusion of a distribution mechanism that transparently allocates returns (interest or revenue sharing) to verified holders. This mechanism features: (i) Pro-rata payout logic via distributeInterest using ERC-20 tokens; (ii) Holder discovery through getHolders to ensure accurate allocation; (iii) Automatic exclusion of non-KYC or frozen accounts; and (iv) Fee distribution (notary, manager, tax) using structured fee management.

This highlights how routine financial obligations—normally handled by registrars, trustees, or fund administrators—can be automated through smart contracts while still meeting regulatory traceability requirements.

4) Full-Cycle Compliance Architecture

Taken together, the system provides a unified compliance-by-design pipeline: (i) KYC registry & role assignment; (ii) Document-bound issuance with dual approval; (iii) Controlled secondary market; (iv) Automated revenue distribution; and (v) Immutable auditability across all phases.

This offers a practical blueprint for tokenized DIRE-like instruments, showing that Indonesia's unique regulatory constraints (notarial authority, OJK oversight, IDR settlement requirements) can be translated into executable on-chain rules. It further demonstrates that tokenization is not limited to digitizing assets, but can restructure governance,

enforcement, and investor protection through smart-contract logic.

5) Implications for Real-World Deployment

Finally, by demonstrating feasibility on the Polygon Amoy testnet and validating all major compliance checks through structured test cases, the study provides a prototype-level reference architecture suitable for: (i) regulatory sandbox trials [18]; (ii) digital-bond and revenue-sharing instruments for SMEs; (iii) DIRE modernization; (iv) hybrid fiat-blockchain integrations; (v) compliant retail-investor participation; (vi) Sharia-compliant implementations; and (vii) integration with Indonesia's emerging e-PPAT digital land-administration ecosystem. Embedding notarial approval and document binding directly into smart-contract logic complements the national shift toward electronic PPAT workflows, enabling verifiable linkage between off-chain land records and on-chain economic-rights issuance.

This contribution bridges theoretical tokenization frameworks with operational, executable compliance mechanisms, offering a step toward standardized digital-asset governance in Indonesia.

B. Limitations of the Current Design

The design focuses exclusively on the issuance stage. While this clarifies the study's contribution, it leaves revenue distribution and secondary trading unaddressed, both of which are essential for the full lifecycle of tokenized real estate [13]. Another limitation is reliance on the ERC1155 standard. Although efficient, it is less integrated into DeFi protocols compared to ERC20, potentially restricting liquidity unless interoperability solutions are developed [16]. The framework also does not provide a detailed KYC/AML mechanism, which remains critical in practice.

C. Exploratory Enhancements

Future enhancements could include Web3Auth integration to simplify investor onboarding while maintaining identity verification, and meta-transaction mechanisms to reduce friction by allowing SPVs or relayers to cover gas fees. These improvements could strengthen both usability and compliance readiness.

VI. CONCLUSION

This study proposed an on-chain issuance framework for real estate tokenization, extending ERC1155 into a compliance-aware design. By encoding dual approval from notaries and fund managers, the system ensures that tokens cannot be minted without legal and financial safeguards. Immutable event logs provide transparency and accountability, aligning blockchain operations with Indonesia's regulatory context.

The framework illustrates that compliance roles can be systematically embedded within smart contracts, offering a practical model for rights-based tokenization in emerging markets. Future research should expand beyond issuance to address revenue distribution, secondary trading, and

integration with DeFi protocols. The urgency now lies in translating such issuance frameworks into pilot implementations, ideally within regulatory sandbox settings, to determine whether tokenization can mature into a credible and scalable financing alternative for Indonesia's property sector.

REFERENCES

- [1] A. Saari, J. Vimpari, and S. Junnila, "Blockchain in real estate: Recent developments and empirical applications," *Land use policy*, vol. 121, p. 106334, Oct. 2022, doi: 10.1016/j.landusepol.2022.106334.
- [2] R10, "Interview with Representative of DIRE operator," Jul. 2025.
- [3] G. of the Republic of Indonesia, *UU 5/1960*. 1960. Accessed: Sep. 25, 2025. [Online]. Available: <https://peraturan.bpk.go.id/Details/51310/uu-no-5-tahun-1960>
- [4] P. P. Indonesia, *PP 24/1997*. 1997. Accessed: Sep. 25, 2025. [Online]. Available: <https://peraturan.bpk.go.id/Details/56273/pp-no-24-tahun-1997>
- [5] R. M. Garcia-Teruel and H. Simón-Moreno, "The digital tokenization of property rights. A comparative perspective," *Computer Law & Security Review*, vol. 41, p. 105543, Jul. 2021, doi: 10.1016/j.clsr.2021.105543.
- [6] BPS, "Suku Bunga Kredit Rupiah Menurut Kelompok Bank, 2025," <https://www.bps.go.id/id/statistics-table/2/MzgZlZl=suku-bunga-kredit-rupiah-menurut-kelompok-bank.html>.
- [7] BPS, "Suku Bunga Deposito Berjangka Bank Umum (Persen), 2025," <https://www.bps.go.id/id/statistics-table/2/MjM2OCMy/suku-bunga-deposito-berjangka-bank-umum-persen-.html>.
- [8] A. A. I. Izadin and R. Yusuf, "Democratizing Real Estate Investment: A Systematic Review of Tokenization in Real Estate," 2024. doi: 10.2139/ssrn.5015941.
- [9] OJK, *OJK 57/2020*. 2020. Accessed: Oct. 02, 2025. [Online]. Available: <https://peraturan.bpk.go.id/Details/227071/peraturan-ojk-no-57poj042020-tahun-2020>
- [10] D., & R. C. Park, "Building infrastructure for bond markets: Lessons from East Asia (ADB Institute Discussion Paper No. 35).," 2006. Accessed: Oct. 03, 2025. [Online]. Available: <https://www.adb.org/sites/default/files/publication/28182/wp035.pdf>
- [11] Y. Chen and C. Bellavitis, "Blockchain disruption and decentralized finance: The rise of decentralized business models," *Journal of Business Venturing Insights*, vol. 13, p. e00151, Jun. 2020, doi: 10.1016/j.jbvi.2019.e00151.
- [12] C. Catalini and J. S. Gans, "Some simple economics of the blockchain," *Commun ACM*, vol. 63, no. 7, pp. 80–90, Jun. 2020, doi: 10.1145/3359552.
- [13] S. Walia, S. Pal, R. Kumra, S. Sarkar, B. Mohanty, and S. Singh, "Exploring Current Research and Future Directions in Commercial Real Estate Investment," *Journal of Real Estate Literature*, pp. 1–29, May 2025, doi: 10.1080/09277544.2025.2496119.
- [14] R. R. R. Chao, J. K. Liu, and K. Su, "Blockchain Based Regulatory Technology Deployment for Real Estate Transaction," *International Conference on Network and System Security*, 2022, pp. 561–579. doi: 10.1007/978-3-031-23020-2_32.
- [15] R. R. KVS, R. Roy, P. Hota, and T. Gupta, "Tokenizing Real Estate in India using Blockchain Technology," in *2024 8th International Conference on Computational System and Information Technology for Sustainable Solutions (CSITSS)*, IEEE, Nov. 2024, pp. 1–4. doi: 10.1109/CSITSS64042.2024.10816882.
- [16] F. Schär, "Decentralized Finance: On Blockchain- and Smart Contract-Based Financial Markets," *Review*, vol. 103, no. 2, 2021, doi: 10.20955/r.103.153-74.[17] R. Indonesia, *UU 7/2011*. 2011. Accessed: Nov. 13, 2025. [Online]. Available: <https://peraturan.bpk.go.id/Details/39182/uu-no-7-tahun-2011>
- [18] OJK, "Peraturan Otoritas Jasa Keuangan Republik Indonesia Nomor 3 Tahun 2024 Tentang Penyelenggaraan Inovasi Teknologi Sektor Keuangan," <https://www.ojk.go.id/id/regulasi/Documents/Pages/POJK-2-2024-Penyelenggaraan-Inovasi-Teknologi-Sektor-Kuangan/POJK%203%20Tahun%202024%20Penyelenggaraan%20Inovasi%20Teknologi%20Sektor%20Keuangan.pdf>.